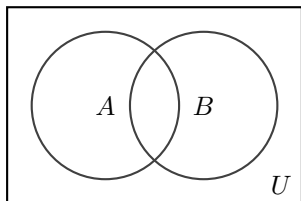


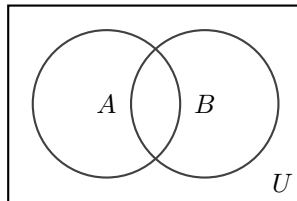
You must show your work to receive credit. Unless stated otherwise, each problem is worth 4 points.

1. [3 points each] Use the provided diagrams to shade in the region corresponding to the indicated set.

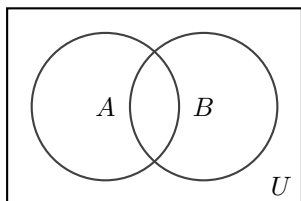
a. $A \cup B$



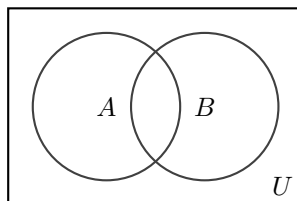
c. A'



b. $A \cap B$



d. $(A \cup B)'$



2. [3 points each] Let $U = \{a, b, c, x, y, z\}$ be the universal set, and $A = \{a, b, c\}$, $B = \{c, x, y\}$, $C = \{y, z\}$ be subsets of U . Find the indicated set.

a. $A \cup B$

b. $A \cap B$

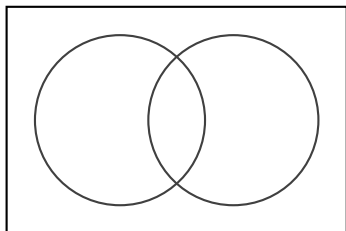
c. $A \cap C$

d. A'

e. $(A \cup B)'$

3. A survey of 1000 students finds that 700 like pizza, 600 like hamburgers, and 400 like both.

Use the provided Venn diagram to label and write the appropriate numbers in the four regions



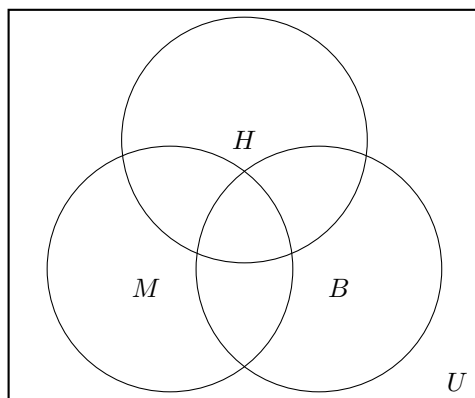
a. How many students like pizza or hamburgers?

b. How many students like pizza but not hamburgers?

c. How many students like neither pizza nor hamburgers?

4. Out of 500 MHU students, 200 are taking a history class, 200 are taking a math class, and 150 are taking a biology class. 50 students are taking all three. 80 students are taking both history and math. 40 students are taking biology but neither math nor history. 90 students are taking both history and biology.

Use the provided Venn diagram to write the appropriate numbers in the eight regions.



- a. How many students are taking only history?

- b. How many students are taking history or Math?

- c. How many students are not taking history, math, or biology?

5. Find the value of each of the following:

a. $\frac{4!}{0!}$

c. $\frac{195!}{193!}$

b. $\frac{27!}{4!25!}$

d. ${}_6C_1$

6. A six-sided dice is thrown, a quarter is flipped, and a card is drawn from a standard 52 card deck. How many different outcomes are possible?

7. There are 10 pigs competing at the county fair in the “that’s some pig” contest.

- a. If the judges see each pig once and only once, how many different ways can they see all the pigs?

- b. The judges will award 1st, 2nd, and 3rd place prizes. How many different ways can the prizes be awarded.

8. A bowling league has seven teams. If every team must play every other team once in the first round of league play, how many games must be scheduled?
9. A 5 player team must be chosen from a group of 8 offensive players and 7 defensive players. How many teams are possible if the team must consist of the following?
- a. 5 offensive players.
 - b. 3 offensive players and 2 defensive players.
 - c. Any mix of offensive and defensive players.
10. Each student at State University has a student ID number consisting of *four* digits (the digits can be repeated) followed by two of the letters *A*, *B*, and *C* (letters cannot be repeated). How many different student numbers are possible?

Honor Statement

On my honor, I have neither given nor received any academic aid or information that would violate the Honor Code of Mars Hill University.

Signature: _____